POLIMI GSoM

Graduate School of Management

Early-Warning System for Risk-Off Regimes

with Domain-Specific Allocation Routing

Unsupervised Anomaly Detection for Regime-Aware Portfolio Allocation

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Problem Statement & Motivation

The gap in traditional risk management:

- Most early-warning systems treat risk-off as a **binary decision**: equity vs. cash.
- But crises are *heterogeneous*: in 2008 the USD was the safe haven (dollar funding squeeze); in 2020 it initially strengthened then weakened; in an inflationary stress regime, **gold** outperforms.
- No existing framework combines **regime detection** with **allocation routing** based on the *nature* of the stress.

Our contribution:

An EWS that answers *two* questions simultaneously:

Investment thesis:

Deploy **1.5**$\times$ **equity leverage** during calm regimes to exploit the equity risk premium, and rely on the EWS to detect stress early enough to **avoid drawdowns** — the key to maximizing the Sharpe ratio.

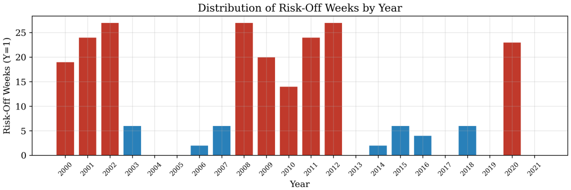
The leverage is justified only if the detection system has sufficient **recall**: missing a crisis at 1.5$\times$ is costlier than at 1.0$\times$.

Key references:

Maggiori (2020) — dollar as global safe asset

Dataset Overview

Source: Bloomberg weekly data
Period: 2000-01-11 to 2021-04-20 (1,111 obs)
Target: $Y = 1 \Rightarrow$ risk-off week (21.3% prevalence)
Missing values: None



Asset Class	#	Examples
Equity (MSCI)	7	MXUS, MXEU, MXJP, MXBR...
FX	3	DXY, GBP, JPY
Commodities	4	Gold, WTI, CRB, Baltic Dry
Bond TR indices	8	US/EU/UK/EM Agg, HY, IG...
Yields/rates	18	US, DE, GB, IT, JP tenors
Other	2	VIX, US Econ Surprise

Risk-off clusters in **2000–02, 2008–09, 2011–12, 2020**. Mid-decade years (2004–05, 2013, 2017) have zero or near-zero positives — this is a data property that critically shapes CV design.

Proxy decisions:

Business Logic: Three Macroeconomic Domains for Routing

USD Domain

The dollar is safe haven when stress is *global but external to the US* or when there is a **dollar funding squeeze** (Avdjiev et al. 2019).

Triggers (signed z-scores):

- $+$ LIBOR-3M spread Δ_{4w}
- $+$ DXY momentum Δ_{4w}
- $+$ VRP (implied $>$ realized)
- $-$ US 10Y Δ change (flight)
- $+$ USA relative vs DM composite

Gold Domain

Gold when stress has a **monetary/inflationary** component or the dollar itself is the problem (Erb & Harvey 2013; Baur & Lucey 2010).

Triggers (signed z-scores):

- $-$ Real yield proxy Δ_{4w}
- $-$ DXY Δ_{4w} (*opposite sign*)
- $+$ JPY strength (safe haven)
- $+$ Equity-bond ρ_{13w} (diversif. failure)
- $+$ Gold-oil ratio Δ_{4w}

MBS Domain

MBS is *not* a crisis refuge — it is **defensive carry** for moderate stress (Diep et al. 2021). Strict activation, conservative by design.

Rule-based activation:

- VIX $\in [20, 28]$ (nervous, not panic)
- Term spread > 0 (normal curve)
- Drawdown $\in [-12\%, -5\%]$

Block if:

- VIX > 30 (acute stress)
- Funding spread $>$ p90
- HY-IG widening $>$ 50bps

Routing Architecture & Decision Matrix

Allocation Decision Matrix

Signal	Condition	Allocation	Weight
Risk-on (0)	—	Equity	$1.5\times$ eq, $-0.5\times$ cash
Risk-off (1)	USD $> \tau_u$ & DXY↑	Cash USD	$1.0\times$ cash
Risk-off (1)	Gold $> \tau_g$	Gold	$1.0\times$ gold
Risk-off (1)	MBS active	MBS	$1.0\times$ MBS
Risk-off (1)	Default	Cash USD	$1.0\times$ cash

Design principles:

- **Priority ordering:** USD $\to$ Gold $\to$ MBS (funding stress is more urgent)
- **Default = Cash:** zero-cost option that doesn't compound errors
- **Transaction costs:** differentiated — equity

Sub-score construction:

For USD and Gold domains:

$$S_d = \frac{1}{|D|} \sum_{f \in D} \text{sign}_f \cdot \frac{x_f - \mu_f^{\text{dev}}}{\sigma_f^{\text{dev}}}$$

where μ, σ are estimated **only on the development set** (no leakage).

Threshold optimization:

Grid search $\tau_u \in \{0.5, \ldots, 2.0\} \times \tau_g \in \{0.5, \ldots, 1.5\}$, selecting by **Calmar ratio** weighted by fold duration on walk-forward CV.

Only *two scalar thresholds* are optimized; the rest of the routing logic is **rule-based from macroeconomic rationale** — this drastically reduces overfitting risk vs. a fully data-driven

Feature Engineering: Stationarity & Cross-Asset Spreads

Stationarity transforms

Anomaly detection models assume a *stable* generating distribution. Non-stationary features violate this assumption fundamentally.

- **Log-returns** $\log(x_t/x_{t-1})$: prices (equity, FX, commodities, bond TR)
- **First differences** Δx_t: yields/rates
- **Level**: VIX, ECSURPUS (already stationary)

Result: 42/42 features pass ADF at 5%.

Collinearity cleanup:

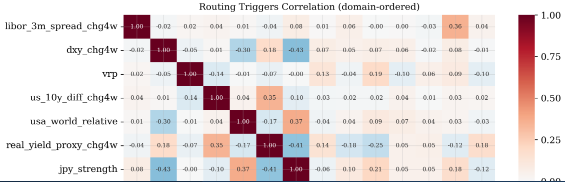
7 features removed ($|\rho| > 0.9$), keeping the more interpretable survivor in each pair.

Final: 56 clean features.

20 cross-asset spreads:

- **Term:** US 10Y−2Y, DE 10Y−2Y, BTP−Bund, US−DE
- **Credit:** HY-IG, EM (log price ratios)
- **VRP:** $\text{VIX} - \hat{\sigma}^{4w}_{\text{realized}}(\text{MXUS}) \cdot \sqrt{52} \cdot 100$
- **Equity-bond rotation:** $R^{4w}_{\text{MXUS}} - R^{4w}_{\text{LUACTRUU}}$
- **Gold-oil ratio:** flight-to-quality vs. growth
- **JPY strength:** $-\Delta_{4w} \log(\text{JPY})$

Each spread: level + 4-week change.



Walk-Forward Cross-Validation with Purged Embargo

Design rationale:

- **Expanding window:** train always starts at t_0, grows forward — mimics real deployment where historical data accumulates
- **4-week embargo:** we drop 4 observations between train end and val start. This eliminates leakage from the 4-week lookback features (Δ_{4w}, rolling windows). **No overlap possible.**
- **Weighted-by-n_{pos} aggregation:** folds 1–2 have 53/64 positives (statistically reliable); folds 3–5 have only 2/10/6 because 2013–2018 is a low-volatility regime. This is a *data property*. Weighting ensures thin folds

Fold	Val Period	Y=1	n_{pos}	Crisis
1	2007–2009	34.9%	**53**	GFC
2	2010–2012	41.7%	**64**	Euro Debt
3	2013–2014	2.0%	2	Taper
4	2015–2016	10.1%	10	China-Oil
5	2017–2018	6.1%	6	Q4 Selloff

Scaler discipline:
`StandardScaler` fit **only on each fold's train**. Test holdout scaled with scaler fit on *entire development set*. Autoencoder early stopping uses a **temporal sub-split of train** — never the validation fold.

This is the #1 source of silent leakage in ML finance papers.

Paradigm: Novelty detection — models see *only normal weeks* ($Y=0$) during training. They learn what "normal" looks like; anything sufficiently different is flagged as risk-off.

1. Multivariate Gaussian (MVG)

Score $= d_M^2(\mathbf{x}) = (\mathbf{x} - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1}(\mathbf{x} - \hat{\boldsymbol{\mu}})$

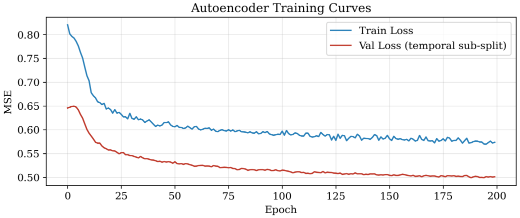
Ledoit-Wolf shrinkage is mandatory: with 56 features and $\sim$360 normals in Fold 1, the sample covariance $\hat{\Sigma}$ is near-singular. Ledoit-Wolf computes the optimal convex combination $\hat{\Sigma}^* = (1-\alpha)\hat{\Sigma} + \alpha \cdot \frac{\mathrm{tr}(\hat{\Sigma})}{p}\mathbf{I}$ that minimizes expected loss under Frobenius norm, producing a well-conditioned estimator.

2. One-Class SVM (RBF)

Maps normals to a high-dimensional feature space via RBF kernel and finds the smallest hypersphere containing them. Score $= -f(\mathbf{x})$. Grid-searched: $\nu \in \{0.05, 0.10, 0.15, 0.22\}$,

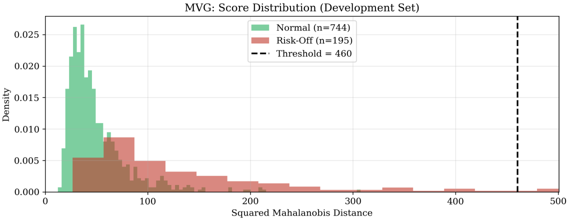
3. Autoencoder

Symmetric bottleneck:
$56 \to 24 \to 12 \to 6 \to 12 \to 24 \to 56$.
ReLU, dropout 0.15, Adam lr=1e-3, MSE loss.
Score $= \frac{1}{p}\|\mathbf{x} - \hat{\mathbf{x}}\|^2$.
Early stopping on **temporal 85/15 sub-split of train normals** (never on CV val — that would leak).



Model Results (Test Holdout 2019–2021)

	F1	AUC-ROC	AUC-PR	Prec	R
MVG	0.516	**0.886**	**0.770**	1.000	0.3
SVM	**0.615**	0.851	0.732	0.750	0.5
IF	0.583	0.810	0.680	0.560	0.6
AE	0.516	0.873	0.764	1.000	0.3



MVG: Mahalanobis distance separates normal (green) from risk-off (red).
Threshold tuned via weighted walk-forward CV.

Key observations:

- MVG and AE: **perfect precision**, low recall — when they flag risk-off they're right, but miss most events
- SVM: best single model on F1 (balanced)
- IF: highest recall (catches 61% of risk-off weeks)
- MVG has highest AUC-ROC (0.886) — the *ranking* is good; the threshold is conservative
- F2 (β=2) weighs recall double: **missing a crisis at 1.5$\times$ is costlier than a false alarm**

Ensemble: Combining Four Diverse Detectors

Why ensemble?

Each model captures different aspects of anomaly: MVG detects distributional shift, SVM finds boundary violations, AE catches reconstruction difficulty, IF identifies structural isolation. If their errors are *uncorrelated*, combining them reduces both FP and FN.

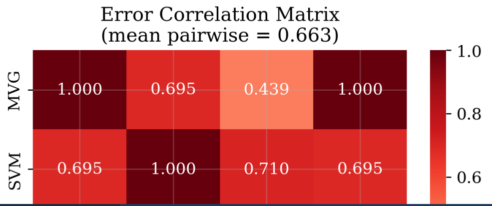
Percentile mapping:

Raw scores are on incomparable scales. We convert each to its **empirical percentile** against the train-normals distribution: $p_i = F_{\text{train}}(s_i)$. This makes all scores $\in [0, 1]$ with comparable semantics.

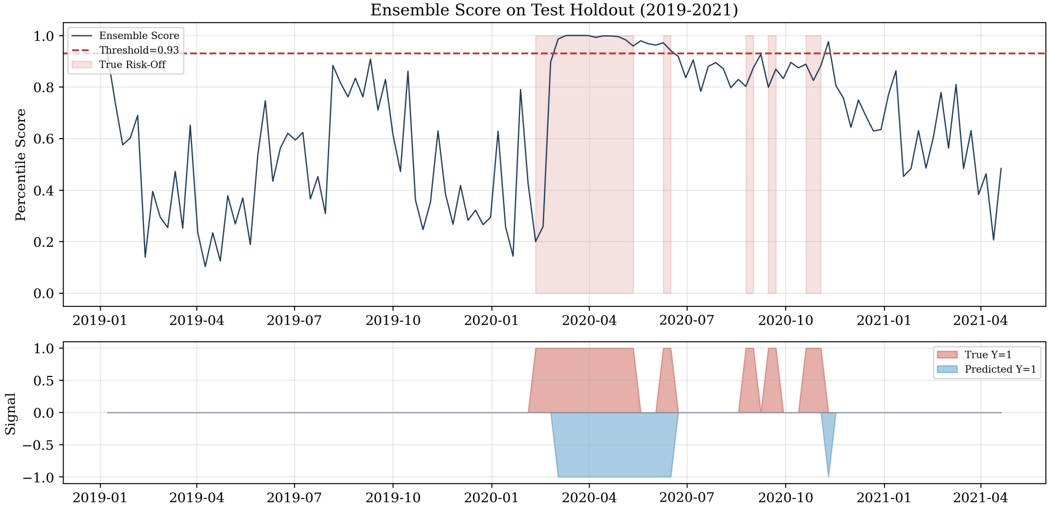
Three variants:

1. **Hard voting:** each model flags 0/1 with own threshold; majority ($\geq 3/4$) wins
2. **Soft mean:** $\bar{p} = \frac{1}{4} \sum p_i$, threshold τ tuned on CV
3. **Soft median:** $\tilde{p} = \text{median}(p_i)$, robust to one rogue

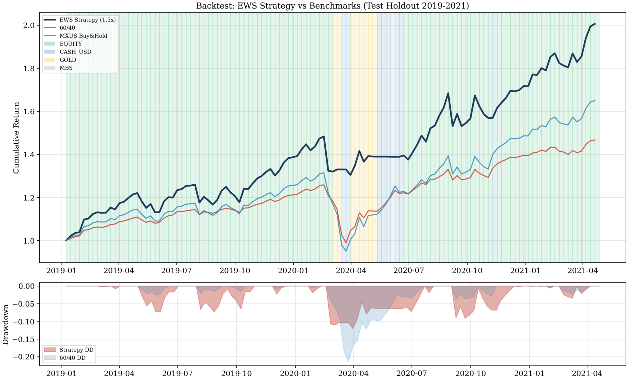
	F1	AUC-ROC	AUC-PR
MVG	0.516	0.886	0.770
SVM	0.615	0.851	0.732
IF	0.583	0.810	0.680
AE	0.516	0.873	0.764
Hard Vote	0.516	0.859	0.742
Soft Mean	**0.650**	0.859	0.742
Soft Median	0.619	0.869	0.759



Ensemble Predictions on Test Holdout



Backtest Results: Strategy vs. Benchmarks (2019–2021)



	Strategy	60/40	B&H
CAGR	**35.2%**	18.1%	24.2%
Sharpe	**1.93**	1.44	1.38
Max DD	**−12.1%**	−21.5%	−27.7%
Calmar	**2.92**	0.84	0.87
Sortino	**2.10**	—	—
Turnover	3.9/yr	—	—

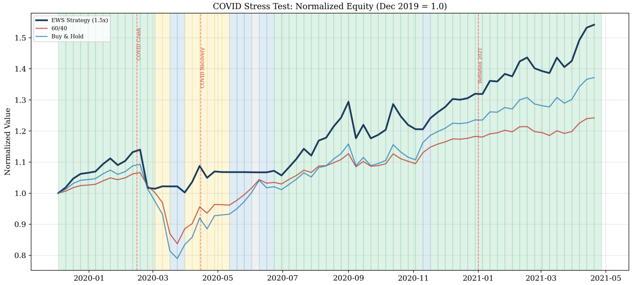
Allocation breakdown:

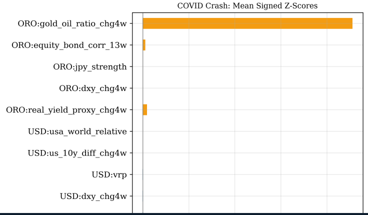
Levered Equity	103 weeks (85.8%)
Gold	8 weeks (6.7%)
Cash USD	8 weeks (6.7%)
MBS	1 week (0.8%)

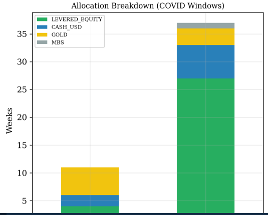
Key insight:

The leverage generates alpha in calm periods (103 weeks at 1.5×), while the EWS limits

COVID Stress Test: Trigger Analysis & Allocation Routing







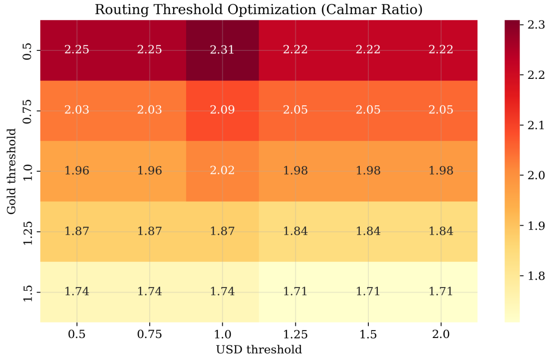
COVID Crash (Feb–Apr 2020):

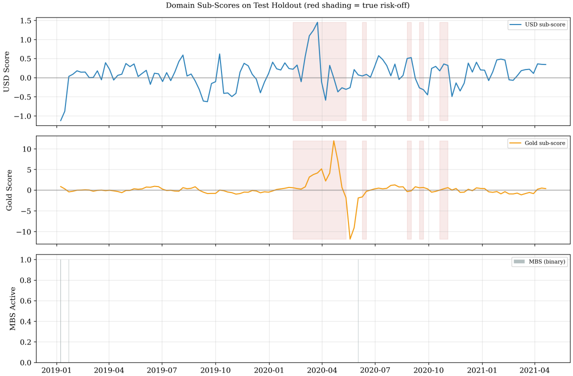
- Ensemble signals risk-off as VIX spikes and equity drops
- **USD triggers fire first**: funding stress (LIBOR-3M spread widens), DXY rallies, VRP explodes — classic dollar squeeze
- Routing sends allocation to **Cash USD and Gold**
- MBS correctly **blocked** (VIX $>$ 30)

COVID Recovery (Apr–Dec 2020):

- Risk-off signals fade as markets stabilize
- System returns to **1.5**$\times$ **equity**, capturing the V-shaped rebound
- Gold score remains elevated (fiscal stimulus $\rightarrow$ inflation expectations)

Threshold Optimization & Sub-Score Dynamics





Grid search over $\tau_{\text{USD}} \times \tau_{\text{Gold}}$. Calmar ratio weighted by fold duration. Optimal: $\tau_u = 1.0$, $\tau_g = 0.5$. The gold threshold is lower because gold signals are less frequent but more persistent.

Top: USD sub-score spikes during COVID crash (funding stress). **Middle:** Gold sub-score elevated during COVID and recovery (stimulus, low real yields). **Bottom:** MBS activates only once (strict conditions).

Limitations, Conclusions & Future Work

Limitations (candidly):

- **Short test period:** 120 weeks, 9 regime switches — results depend on few trades; not yet statistically robust
- **Thin CV folds:** Folds 3–5 have 2–10 positives; n_{pos}-weighting mitigates but doesn't eliminate noise
- **Proxy constraints:** MXUS $\neq$ MSCI World; log-ratios $\neq$ credit spreads
- **Weekly frequency:** misses intra-week volatility events
- **No 2022 data:** the inflation/rate shock regime — the true test for Gold routing — is absent
- **Leverage risk:** at 1.5×, a *missed* crisis costs more; the system's 35% recall gap is a genuine concern

Conclusions:

- The **ensemble of unsupervised detectors** outperforms all single models (F1 0.65 vs. 0.62 best single)
- **Domain-specific routing** adds genuine intelligence vs. binary risk-on/off
- **Walk-forward + embargo** provides methodologically sound OOS evaluation
- Strategy achieves **Sharpe 1.93** with **Max DD −12%** vs. −28% for buy&hold

Future work:

- Daily frequency for finer granularity
- SHAP / Mahalanobis decomposition for signal attribution
- Include 2022–2025 data (inflation regime)